CS2-XX: Advance Database Management System

General Information

Course Number	CS2-XX
Credit Hours	2+1 (Theory Credit Hour = 2, Lab Credit Hours = 1)
Prerequisite	CS-X001 – Advance Database Management System
Course Coordinator	
Consultation Hours	

Course Objectives

The aim of the course is to enhance the previous knowledge of database systems by deepening the understanding of the theoretical and practical aspects of the database technologies, and showing the need for distributed database technology to tackle deficiencies of the centralized database systems.

Moreover, it focuses to introduce the basic principles and implementation techniques of distributed database systems, and expose emerging research issues in database systems and application development.

Catalog Description

CS2-XX

Student Assessment during semester

S.No:	Assessment Task (examination, quizzes, assignments, presentations, etc.)	Proportion of Final Assessment/Marks
1	Final exam	50
2	Mid-Exam	30
3	Quizzes/Assignment	10
4	Presentation/Class Participation/Attendance	5
5	Class Project	5

Course Outline

Session No.	Week No.	Topics	Assignments/ Presentations/Quizzes	Suggested Readings (Books)
01-03	1-2	• Introduction • Overview of basic databases • Introduction to Data Models such as Relational, Object relational, object oriented, and Spatial Databases		Text Book 1
04-06	2-3	• File Organization Concepts and Transactional Processing		Text Book 1
07-08	3-4	• Transactional Processing and Concurrency Control Techniques		Text Book 1
09-10	5	• Recovery Techniques, Query Processing and Optimization	Quiz - 1	Text Book 1
11-12	6	• Integrity and Security • Database Administration (Role Management, Managing Database Access, Views)		Text Book 1
13-14	7	• Distributed Database Systems	Assignment - 1	Text Book 1
15-16	8	• Emerging Trends in Database Systems		Text Book 1
		Mid Exam		
17-20	11-12	• XML Schema • Introduction to MongoDB • **Project Discussion and Idea Finalization**		Text Book 1
21-22	13	• NoSQL		Text Book 1
23-26	14-15	• Spatial DB • Spatial Indexing Structure • Multimedia DBs	Assignment - 2	Text Book 1
27-30	16-17	• Multidimensional DBs • Big Data	Quiz - 2	Text Book 1

	18	**Project Presentations**	**Project**	
Final Exams				

Text Book

1. Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition by Thomas Connolly and Carolyn Begg
2. Database Management Systems, 3rd Edition by Raghu Ramakrishnan, Johannes Gehrk

Reference Material

1. Database System Concepts, 6th Edition by Avi Silberschatz, Henry F. Korth and S. Sudarshan
2. Database Systems: The Complete Book, 2nd Edition by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom

Course Learning Outcomes

	Course Learning Outcomes (CLO)	Bloom (Taxonomy)
1	Understanding advance data models, technologies and approaches for building distributed database systems	C2 (Understand)
2	Applying the models and approaches in order to become enabled to select and apply appropriate methods for a particular case	C3 (Apply)
3	To develop a database solution for a given scenario/ challenging problem in the domain of distributed database system	C3 (Apply)

CLO-GA Map

	GAs									
CLO ID	**GA1**	**GA2**	**GA3**	**GA4**	**GA5**	**GA6**	**GA7**	**GA8**	**GA9**	**GA10**
CLO 1	1	0	0	0	0	0	0	0	0	0
CLO 2	0	1	0	0	0	0	0	0	0	0
CLO 3	0	0	1	0	0	0	0	0	0	0

Approvals

Prepared By	
Approved By	Not Specified
Last Update	